

Jev in Practice: A Composable Python Toolkit for TypeSafe's System One Decision Model

Batching Speedups, Confidence-Gated Routing, and Calibrated Decisions over a Single HTTP Endpoint

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1 Abstract

Large language models are usually consumed as text generators, yet a growing class of applications needs them as *decision components*: bounded, typed answers that software can branch on. This manuscript presents `daf-jev`, a composable Python toolkit for the TypeSafe Jev (System One) decision model [TypeSafe AI, 2026d,e]. The package reduces the entire decision surface to three typed question primitives — `noul` (a calibrated yes/no), `choice` (a labelled pick with a full probability distribution and a confidence), and `score` (a rating over ordered levels with a probability-weighted expected value) — all carried by a single HTTP endpoint. Pure-logic composition patterns (`composite_score`, `confidence_gate`, `route/pick`) turn typed answers into executable decisions with no additional network traffic, and a batch `Evaluator` fans a fixed question set out over many states through either the synchronous or the asynchronous client. Two live-API benchmarks quantify the practical payoff of this design. Batching a batch’s worth of questions into one call yields wall-time speedups of 3.98, 8.77, and 18.55 across the configured batch widths, with a token-cost ratio of 4.22 at the widest setting; end-to-end decision pipelines (network round trip plus composition logic) complete at a median wall time of 0.133 for composite scoring and 0.129 for intent routing against the `jev-latest` model. Decision quality is validated as well as speed: a self-consistency calibration benchmark reports an expected calibration error of 0.0730 and a Brier score of 0.0252 over $6 \text{ states} \times 5 \text{ repeats}$ against the `jev-latest` model, and the same core is exposed to non-Python consumers through an MCP server (`daf-jev serve`), an agent skill, and a set of runnable examples. Every numeric value in this manuscript is generated from the same analysis outputs as the figures and tables, so the prose cannot drift from the data.

Keywords: decision models, LLM APIs, calibrated confidence, batching, confidence-gated routing, Python client, reproducible research

2 Introduction

Software that consumes large language models faces a structural mismatch. Language models emit prose; software needs values. When an application must branch — route a support ticket, gate a refund, escalate a low-confidence judgment — the prevailing pattern is to ask a generative model for free-form text and then recover a decision from that text with parsers, regular expressions, and hope. Every such recovery step is a silent failure surface: the model can hedge, reformat, or refuse, and the calling code has no contract that tells it when the answer is trustworthy.

TypeSafe’s System One model attacks the mismatch at the source. Rather than generating text, a decision model returns typed answers — probabilities, labelled picks, and confidence values that conform to the shapes the calling code already expects [TypeSafe AI, 2026d]. Its training path, reinforcement learning for calibrated decisions (RLCD), optimizes for decisions and calibrated probabilities rather than fluent generations, so that uncertainty is expressed through well-calibrated probability distributions instead of overconfident verdicts [TypeSafe AI, 2026e]. The model is positioned explicitly as a component for *AI-powered software*, not an agent: code owns the control flow, and the model supplies narrow, structured judgments embedded at exactly the points where the software needs programmable common sense.

2.1 The problem this package solves

The System One decision surface is exposed through one HTTP endpoint and three question types. Using it directly from Python means hand-writing request payloads, discriminating answer unions, mapping HTTP status codes onto exceptions, retrying transient failures, and re-serializing the same state once per question. None of that is research; all of it is boilerplate with real failure modes. Worse, the *composition* layer — turning a set of typed answers into an actual software decision — lives in cookbook prose rather than in tested code.

daf-jev closes that gap. It is a small, dependency-light Python package that provides:

- **Typed primitives** — ergonomic builders (`noul()`, `choice()`, `score()`) and a `QuestionSet` container for the three question types, with strict wire-shape validation on both the request and the answer side.
- **A robust client** — `JevClient` and `AsyncJevClient` wrapping the single System One endpoint, with a pure, injectable retry policy for transient server-side conditions, a complete typed exception hierarchy, and a models listing.
- **Composable decision patterns** — pure functions over answers (`composite_score`, `confidence_gate`, `route`, `pick`) implementing the documented TypeSafe patterns, plus a concurrent batch `Evaluator` that runs a fixed question set over many states without ever aborting the batch.

The package follows the template conventions of the surrounding research infrastructure: logic lives under `src/daf_jev/`, orchestration stays in thin scripts, and every dynamic fact that reaches this manuscript flows through generated variables rather than hand transcription. Third-party efforts in the same direction — an architectural registry for Jev-style decision layers [Claburn, 2026], an orchestration router over the same endpoint family [Corvin, 2026], and a standalone evaluation harness [Morales, 2026] — share the motivation but not the scope; positioning against them is deferred to sec. 8.

2.2 Reader’s guide

sec. 3 gives a technical account of the Jev (System One) decision model itself — the answer shapes, the parallel evaluation semantics, the latency envelope, and the calibration guarantees — distinguishing primary documentation claims from third-party characterizations. sec. 4 describes the package architecture: its layers, its primitives, its composition patterns, and the design rulings that shaped them. sec. 5 reports the measured batching speedups, token-cost ratios, and decision-pipeline latencies, all bound to figures and tables generated from the benchmark outputs. sec. 6 records the exact environment and configuration the measurements ran under, sec. 7 certifies how every artifact can be regenerated and verified, and sec. 8 closes with limitations, related work, and positioning.

3 The Jev (System One) Decision Model

This section states the model-level facts that `daf-jev` builds on. Throughout, claims taken from the primary `TypeSafe` documentation snapshot bundled with this repository are attributed to it; characterizations drawn from independent projects are marked as third-party and carry their own citations.

3.1 A model class, not a chat endpoint

System One is `TypeSafe`’s model for building *AI-powered software* rather than agents. It does not generate code, plan, or choose its own next action; it returns narrow, structured decisions over unstructured input, so that control flow, deterministic rules, and side effects remain entirely in application code [[TypeSafe AI, 2026d](#)]. The documentation frames this as a third architecture beside traditional software (complex decision trees of reliable primitives) and LLM agents (loops that accumulate off-the-rails opportunities): in AI-powered software, the model appears only where the system needs programmable common sense, and each AI task is atomic and constrained [[TypeSafe AI, 2026d](#)].

The training path behind this behavior is RLCD — reinforcement learning for calibrated decisions. Where RLHF and RLVR adapt a pretrained model to produce better *text*, RLCD optimizes for a different output contract: the model does not generate prose at all, it returns decisions and probabilities, communicating uncertainty through calibrated distributions instead of tending toward overconfidence [[TypeSafe AI, 2026e](#)]. Independent projects have adopted the same framing for their own decision layers — an architectural registry of Jev-style components [[Claburn, 2026](#)], a routing layer over the endpoint family [[Corvin, 2026](#)], and an evaluation-oriented harness [[Morales, 2026](#)] — and their descriptions are consistent with, but not authoritative for, the primary contract.

3.2 One endpoint, three question primitives

Every System One interaction is a single request to one decision endpoint carrying a *state* (the context to judge) and a dictionary of questions keyed by caller-chosen identifiers [[TypeSafe AI, 2026a](#)]. Each question is one of three primitives, discriminated by its `type` field:

- **noul** — a yes/no judgment with optional contrasting criteria for the true and false readings. The answer is a single calibrated probability on the unit interval, where the endpoints mean “no” and “yes” respectively.
- **choice** — a pick among named options, each with an optional free-text description. The answer carries the selected label, a probability distribution over *all* options (summing to unity), and a scalar confidence.
- **score** — a rating over an ordered list of level descriptions (at least two). The answer carries a probability-weighted score that may be fractional, the ordered legend of level descriptions, the probability distribution over levels, and a scalar confidence.

The response pairs the answers with a usage record (input and output token counts) and an identifier of record returned in the response headers, which clients should preserve for tracing. Error conditions are enumerated by status class — authentication, permission, not-found, bad request, validation, rate limiting, overload, and internal faults — and the transient classes among them are the ones a client is expected to retry.

3.3 Parallel, composable evaluation semantics

Two semantic properties of the model make composition safe, and both are primary-source guarantees rather than implementation accidents [[TypeSafe AI, 2026d](#)]:

1. **Independence by construction.** Questions are evaluated independently and in parallel; one primitive’s result never becomes hidden context that shifts another primitive’s answer. Composition logic therefore cannot create hidden coupling by *asking* more questions — coupling enters only when code combines answers, where it is visible and testable.
2. **Comparability.** Outputs are sortable numeric values that can drive thresholds, comparisons, and branching directly, without an intermediate natural-language interpretation step.

Because questions are independent, any number of them can be batched into a single call: the endpoint evaluates them as a parallel sampler, and the documented parallel-questions pattern recommends exactly this for workloads that would otherwise repeat the same state in many single-question requests [[TypeSafe AI, 2026c](#)]. Batching changes the cost structure (one round trip instead of many, one state transmission instead of a repetition per question) but not the answers themselves — an invariant our benchmark in sec. 5 relies on and verifies.

3.4 The latency envelope and calibration

The primary documentation describes System One as fast enough for real-time request paths and user interfaces, with most queries completing on a millisecond-scale latency envelope [[TypeSafe AI, 2026d](#)]. That envelope is what allows the composition patterns in sec. 4 to run *inline* — a confidence gate or an intent route is cheap enough to sit on an interactive request path rather than in a background queue.

Calibration is the second load-bearing property. Because RLCD trains the model to express uncertainty as calibrated probability mass rather than as confident-sounding text, the `confidence` field on `choice` and `score` answers, and the probability distributions beside

them, are meaningful inputs to policy decisions [TypeSafe AI, 2026e,b]. The documented confidence-routing pattern treats confidence as a *second decision axis* alongside the primary answer value: high-confidence answers can be automated, mid-band answers routed to review, and low-confidence answers escalated to a human or a fallback policy [TypeSafe AI, 2026b,c]. daf-jev implements that pattern literally in `confidence_gate` and `route` (sec. 4), and sec. 5 measures how much latency that policy layer adds on top of a network round trip — in practice none beyond measurement noise, since the composition logic is pure computation.

4 Methodology: Package Architecture

daf-jev is organized as a strict layering: an ergonomic surface (CLI, scripts, benchmarks) on top of pure-logic composition, on top of primitives and wire types, on top of a single transport and client layer that owns all I/O. fig. 1 shows the module graph. The layering is enforced by convention: the composition and primitive modules import no I/O machinery, so every decision pattern is testable against constructed answers without a network stand-in.

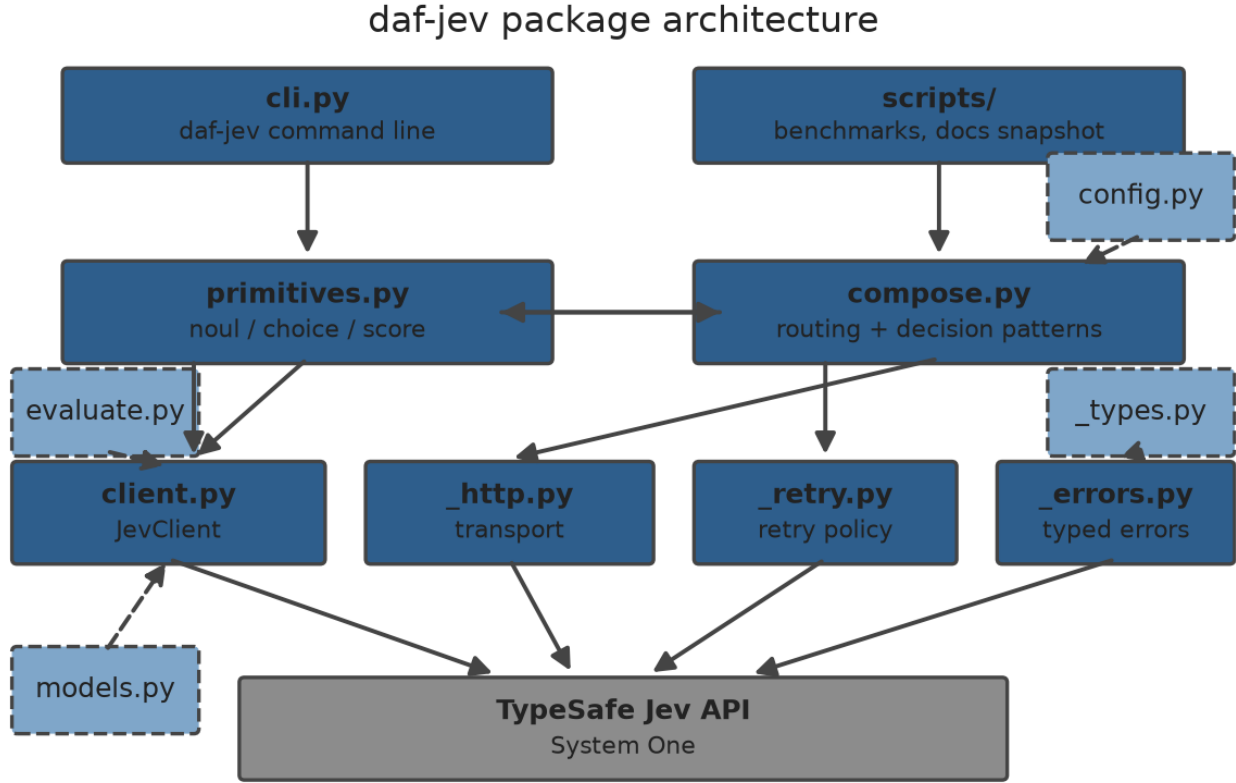


Figure 1: Package layer diagram for daf-jev. The CLI (`src/daf_jev/cli.py`) and the thin orchestration scripts (`scripts/`, `benchmarks/`) sit on the application layer; `compose.py` (pure decision patterns) and `primitives.py` (typed question builders and `QuestionSet`) form the logic layer over the wire dataclasses in `_types.py`; `client.py` and `_http.py` own the single transport to the System One endpoint, with `_retry.py` and `_errors.py` providing the pure retry policy and the typed exception hierarchy; `models.py`, `config.py`, and `evaluate.py` enter as side inputs and the batch harness respectively. Boxes are named modules only; no measured values appear in the diagram.

4.1 Layered architecture

The layers, bottom-up:

- **Transport** (`_http.py`) — a minimal Transport protocol with a synchronous and an asynchronous `httpx` implementation. It knows nothing about decisions; it posts JSON to a path and returns the raw response.
- **Client** (`client.py`) — `JevClient` / `AsyncJevClient` resolve the API key from the environment (injected mapping, process environment, then a `.env` file, in that precedence order via `config.py`), build the request for a state plus a mapping of questions, apply the retry policy, and map HTTP status classes onto the typed exception hierarchy of `_errors.py`. The retry policy itself (`_retry.py`) is a *pure* function of the attempt number and an optional server-supplied retry hint — exponential backoff with a base, a cap, and jitter, with the hint winning when present — so its timing behavior is unit-testable without sleeping.
- **Wire types** (`_types.py`) — frozen dataclasses for the three question types and the three answer types, a strict parser that rejects unknown answer shapes, a usage record, and the top-level response object with cached per-type views (`nouls`, `choices`, `scores`).
- **Primitives** (`primitives.py`) — the ergonomic builders `noul()`, `choice()`, `score()` and the `QuestionSet` mapping container with additive composition (`add`, `merge`, `to_wire`). No I/O.
- **Composition** (`compose.py`) — pure functions over answers, described below. No I/O; network access happens only through a client injected by the caller for multi-call helpers.
- **Harness** (`evaluate.py`) — the batch Evaluator, described below.

Jev question primitives and typed answers

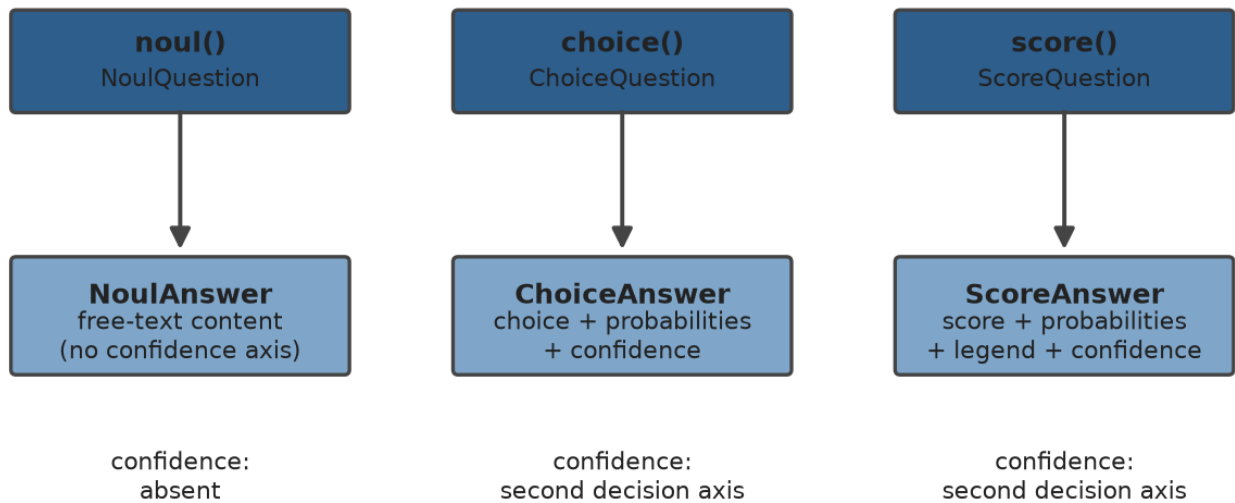


Figure 2: The three question primitives of the System One surface and the typed answer shapes daf-jev parses them into. A `noul` question yields a calibrated yes/no probability; a `choice` question yields a selected label, a full probability distribution over the named options, and a scalar confidence; a `score` question yields a probability-weighted score over ordered levels together with the level legend, the level distribution, and a confidence. The diagram is a schematic of shapes only — it carries no measured data.

4.2 Typed primitives

The builders in fig. 2 map one-to-one onto the wire contract of sec. 3. Validation is strict and eager: score criteria must carry at least two ordered levels, choice criteria must be non-empty, and the answer-side parser rejects any payload whose shape does not match the discriminated union exactly. The payoff of strictness is that downstream composition code can pattern-match on answers without defensive parsing — the type system, not runtime guesswork, guarantees the shapes.

4.3 Composable decision patterns

The composition layer implements the documented TypeSafe patterns as pure functions [TypeSafe AI, 2026c]:

- **composite_score(answer, weights=None)** — the expected value of a score answer over its level indices, uniform by default or under caller-supplied weights (which must be finite and positive-massing). This turns an ordinal rating into a comparable scalar without a second model call.
- **confidence_gate(answer, threshold, below)** — returns the answer’s primary value when its confidence meets the caller’s threshold, and the below verdict otherwise. The threshold is owned by the calling code, never by the package.
- **route(answer, handlers, min_confidence, fallback)** — dispatches to the handler registered for a choice answer’s label, gated on a minimum confidence with an optional fallback handler; pick fans the same dispatch across every answer in a mapping.

Batching fan-out and the Evaluator. Because the model evaluates questions independently and in parallel (sec. 3), the cheapest correct strategy is to batch a fixed question set into one call. The `Evaluator` in `evaluate.py` inverts the axis: it holds the question set fixed and runs it over *many states*, concurrently — through a thread pool for the synchronous client or `asyncio` with a semaphore for the asynchronous client. Per-state failures never abort the batch; they are captured into the evaluation record alongside the answers and the measured latency, so one malformed state cannot cost the run. This harness is what the benchmark scripts in sec. 5 reuse for their fan-out measurements.

Confidence-gated routing. fig. 3 illustrates the three-band policy that `confidence_gate` and `route` implement: answers whose confidence clears the upper band are automated, mid-band answers are routed to review, and answers below the lower band are escalated. The boundary markers in the figure are *example thresholds* — the semantics are the bands, not any particular cut points, which remain policy owned by the caller.

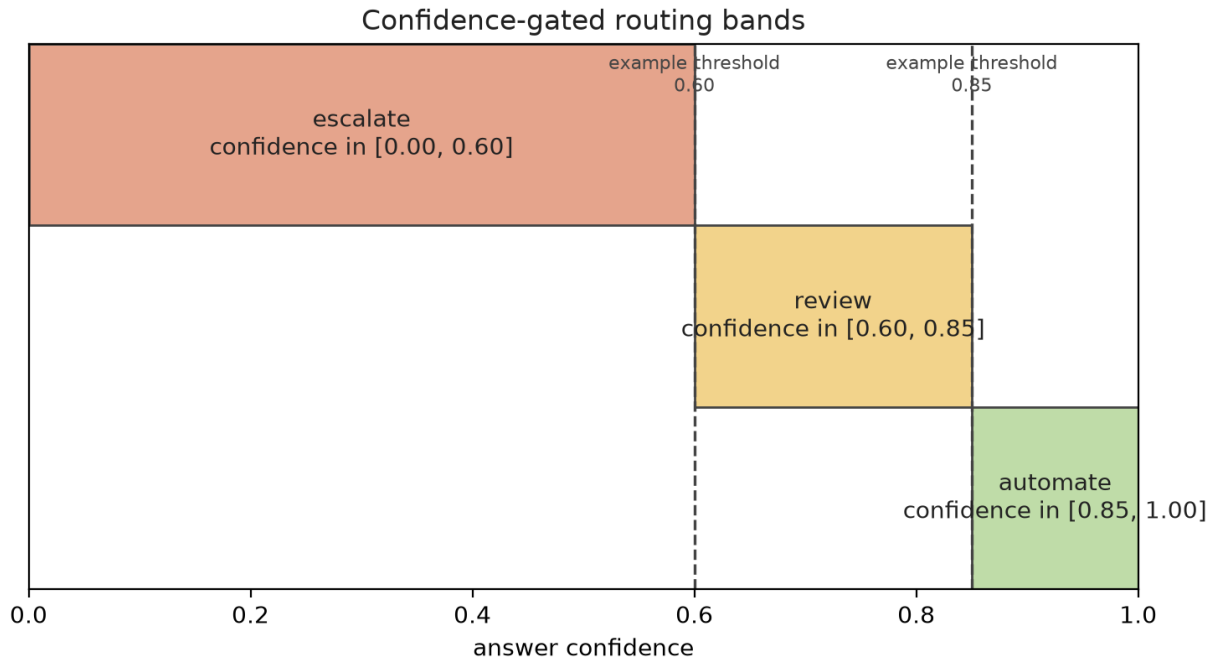


Figure 3: Parametric illustration of confidence-gated routing as implemented by `confidence_gate()` and `route()` in `src/daf_jev/compose.py`. The horizontal axis is the model-reported confidence on the unit interval; the three horizontal bands assign an action per confidence region — automate, review, escalate. The boundary markers are annotated as example thresholds: they illustrate the band semantics and are not fitted or recommended values.

4.4 Design rulings

Four rulings shaped the codebase and are worth stating as contracts:

1. **Pure logic, injected I/O.** Everything under `compose.py`, `primitives.py`, `_types.py`, and `_retry.py` performs no I/O; network access is confined to the transport and client layer and is injectable (clock, sleep, transport) for deterministic tests.

2. **Real stand-ins, no mocks.** The test suite drives the real HTTP transport against a real local HTTP server fixture rather than patching client internals, so retry, timeout, and error-mapping behavior is exercised end to end.
3. **Pure retry policy.** Backoff timing is a function of the attempt number and the server’s retry hint alone — testable without sleeping, and identical across the sync and async clients.
4. **Snapshotted documentation.** The bundled documentation snapshot carries a manifest of per-page content hashes and a snapshot identifier, and a CLI subcommand re-hashes the tree to detect drift (sec. 7). Prose claims about the model are therefore checkable against an immutable, verified source rather than a moving website.

4.5 Integration surfaces

The Python API is the primary door into the package, but not the only one. Three additional surfaces sit on top of the same layering — transport, client, primitives, composition — and add accessibility rather than logic: each one delegates to the very modules described above and contains no decision code of its own.

- **MCP server** (`src/daf-jev/mcp_server.py`, launched with `daf-jev serve` over the stdio transport). It exposes the core to any MCP-capable agent host through tools that mirror the package one-to-one: `jev_ask` and `jev_evaluate` drive the client’s single-call and batch paths, `jev_models` reports the available model surface, `jev_composite_score`, `jev_confidence_gate`, and `jev_tiered_gate` expose the composition patterns, and `jev_docs_verify` re-runs the documentation drift check of Design ruling 4; a `jev://docs/snapshot` resource serves the bundled documentation snapshot itself. The server is a thin adapter — it owns no retry policy, no parsing, and no thresholds.
- **Agent skill** (`skills/daf-jev/SKILL.md`). A skill manifest that tells a coding agent when and how to reach for the CLI subcommands and composition helpers, so an agent can use the package correctly without reading its source.
- **Runnable examples** (`examples/`). Four self-contained scripts — `quickstart.py`, `triage_router.py`, `composite_scoring.py`, and `evaluate_corpus.py` — walk the typical integration path from a first call to a batch evaluation, doubling as executable documentation of the composition layer.

None of these surfaces introduces a second implementation of the model semantics: the MCP tools call the same client and compose functions the CLI calls, the skill documents those entry points, and the examples exercise them. A behaviour fixed in the composition layer therefore reaches every consumer at once.

5 Results

This section reports the three live-API benchmarks that quantify the design claims of sec. 4: the batching benchmark, which reproduces the documented parallel-questions pattern [TypeSafe AI, 2026c], the decision-pattern latency benchmark, and the self-consistency calibration benchmark. All run against the real jev-latest model; every value below is injected from the benchmark outputs at render time, and the figures are regenerated from the same JSON files, so prose, tables, and figures share one source of truth. The batching and latency results reported here were recorded on 2026-09-16; the calibration results on 2026-09-16.

5.1 Batching speedup and token cost

The batching benchmark compares two strategies over the same state and question mix: one call carrying all questions in a batch, versus one sequential single-question call per question, repeated for each configured batch width over multiple measured runs. The answers are unchanged by batching (the parallel-sampler semantics of sec. 3); what changes is wall time — a single round trip versus one round trip per question — and token cost, because the sequential strategy re-sends the state once per question.

fig. 4 shows the measured speedup per batch width together with the token-cost ratio on a secondary axis.

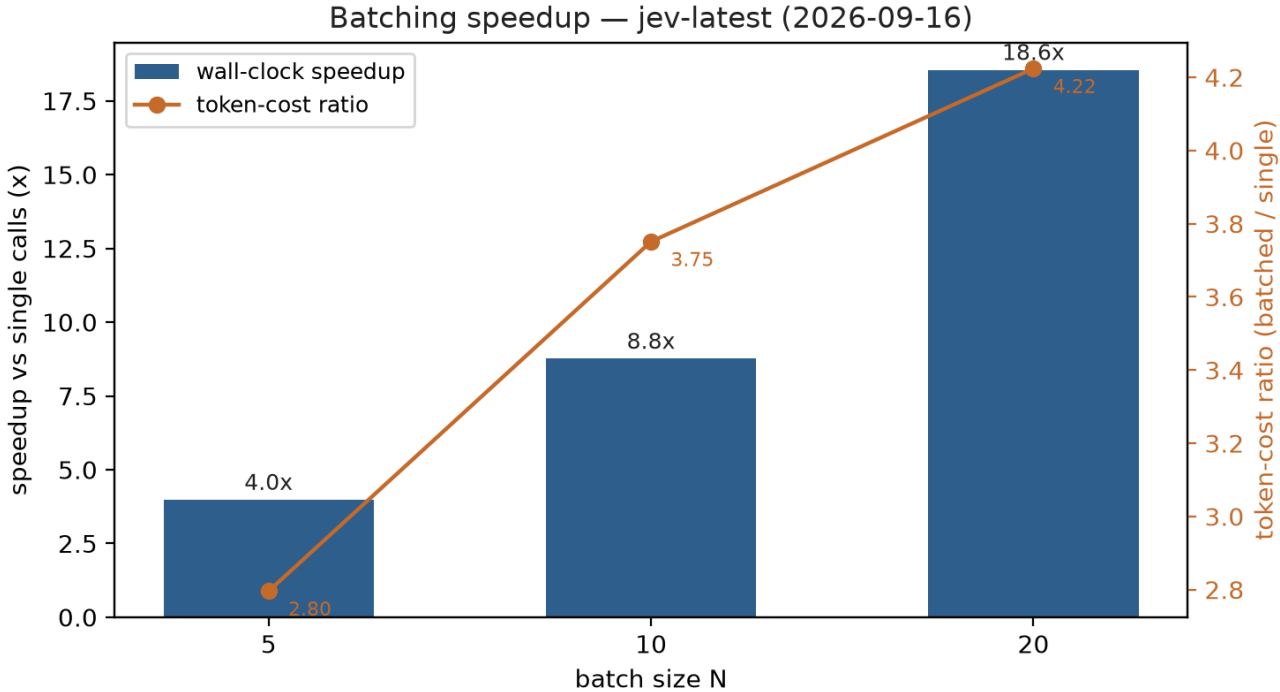


Figure 4: Batching speedup of the jev-latest model versus sequential single-question calls, measured by `benchmarks/bench_batching.py` on 2026-09-16. Bars give the wall-time speedup of one batched call over one sequential call per question for each configured batch width (5, 10, 20 questions, values in tbl. 1); the secondary axis shows the token-cost ratio, which falls below unity because the sequential strategy re-sends the state once per question. Bars are annotated with their values; the title in the rendered figure carries the model and run date read from the benchmark JSON itself.

tbl. 1 tabulates the measured speedups. The speedup grows with batch width, as expected from the round-trip accounting: the fixed per-call overhead is amortized over more questions, and the state is transmitted once rather than once per question.

Table 1: Wall-time speedup of one batched call versus one sequential call per question, per configured batch width, recorded by `benchmarks/bench_batching.py` against the jev-latest model on 2026-09-16. Values are injected from the benchmark JSON at render time.

Questions per batch	Wall-time speedup vs sequential
5	3.98
10	8.77
20	18.55

Token cost moves in the same direction. At the widest configured batch width the batched strategy consumes a token ratio of 4.22 relative to the sequential strategy: the state paragraph dominates the input tokens of a single-question call, so re-sending it per question makes the sequential strategy strictly more expensive in addition to being slower.

5.2 Decision-pipeline latency

The second benchmark measures end-to-end latency of two composition pipelines — the network call plus the local composition logic, exactly as an application would run them:

- **composite_score** — one score call, then `composite_score` over the answer, then a `confidence_gate` verdict;
- **intent_routing** — one choice call, then `route()` dispatching to a trivial handler.

Each pipeline is executed 6 times; [tbl. 2](#) reports the median (p50) and tail (p95) wall times per pipeline, and [fig. 5](#) plots them side by side.

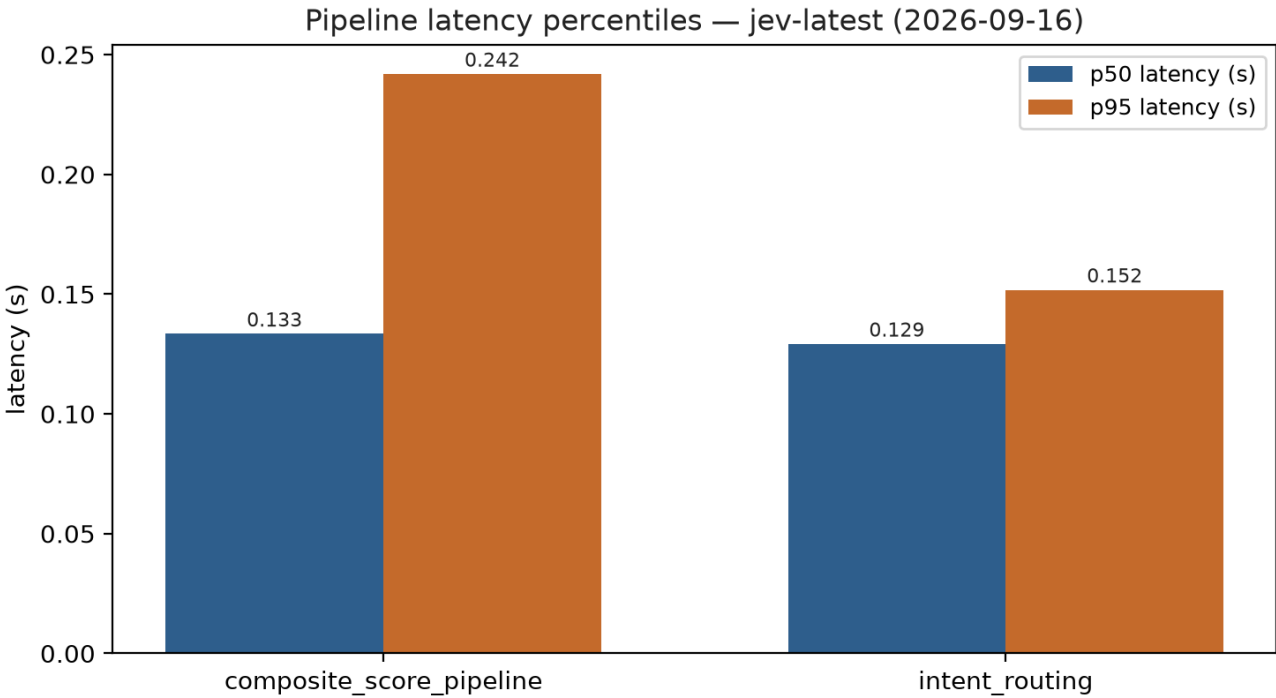


Figure 5: Median (p50) and tail (p95) end-to-end wall time per decision pipeline, measured by `benchmarks/bench_patterns.py` over 6 runs against the `jev-latest` model on 2026-09-16. Each bar is one pipeline (composite scoring, intent routing) with paired p50/p95 groups; axis labels are read from the benchmark JSON fields at figure-generation time. Values are tabulated in [tbl. 2](#).

Table 2: End-to-end wall time (network round trip plus local composition logic) per decision pipeline, median and tail over 6 runs recorded by `benchmarks/bench_patterns.py` against the `jev-latest` model on 2026-09-16. Values are injected from the benchmark JSON at render time.

Pipeline	Median wall time, p50 (s)	Tail wall time, p95 (s)
composite_score	0.133	0.242
intent_routing	0.129	0.152

5.3 Calibration

Latency says nothing about whether the composed decisions are trustworthy, so the third benchmark measures how far the model’s reported confidence tracks its own agreement behaviour. `benchmarks/bench_calibration.py` runs 6 short states 5 times each against the `jev-latest` model and scores choice answers under a *self-consistency proxy*: a sample counts as correct when it agrees with the modal choice across the repeats of the same state. Because a yes/no probability carries no per-sample label to agree with, noul answers are scored for repeat-to-repeat stability instead, via the mean absolute gap between every pair of repeated answers. No external ground truth is consulted anywhere in this benchmark — the correctness proxy is self-agreement, not verified outcomes.

[tbl. 3](#) reports the aggregate scores over the run, and [fig. 6](#) plots the reliability curve per confidence bucket.

Table 3: Calibration summary over 6 states $\times$ 5 repeats recorded by `benchmarks/bench_calibration.py` against the `jev-latest` model on 2026-09-16. Choice correctness is a self-consistency proxy — agreement with the modal choice across repeats — not accuracy against ground truth; the mean pairwise noul gap measures repeat-to-repeat answer stability rather than correctness. Values are injected from the benchmark JSON at render time.

Metric	Value
Expected calibration error (choice, proxy)	0.0730
Brier score (choice, proxy)	0.0252
Mean pairwise noul gap	0.0050

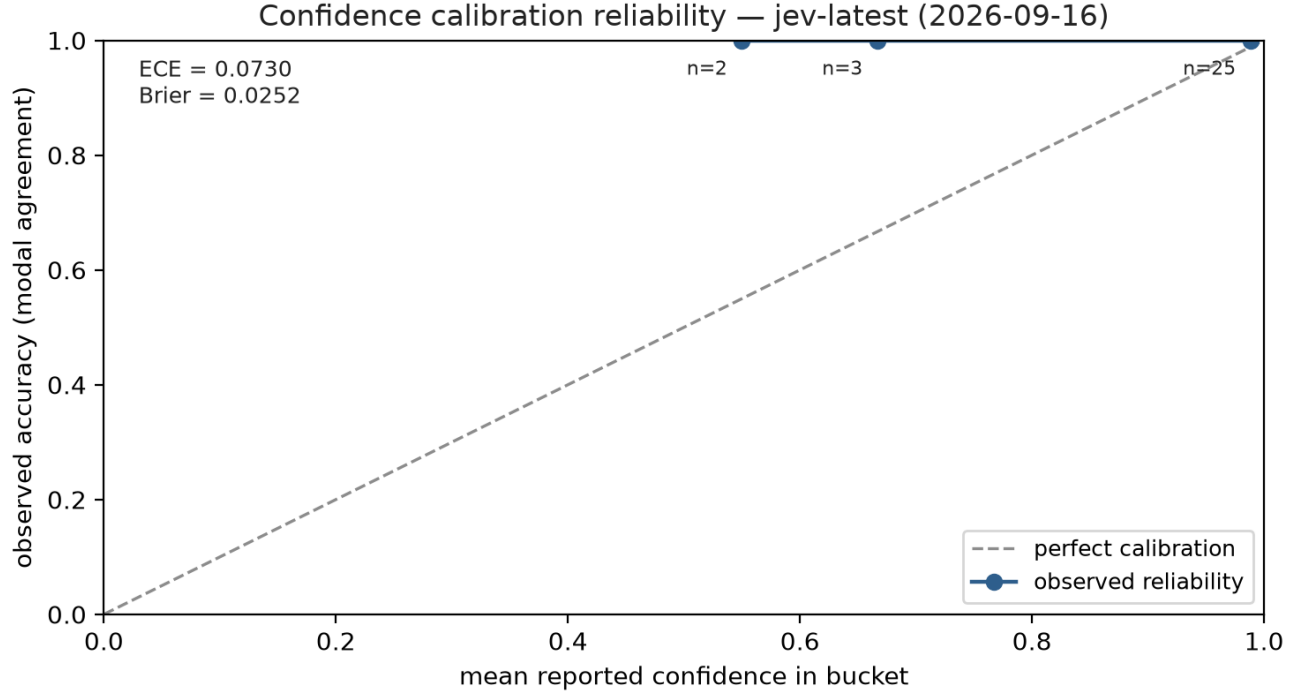


Figure 6: Reliability diagram of choice-answer confidence under the self-consistency proxy, measured by `benchmarks/bench_calibration.py` over 6 states $\times$ 5 repeats against the `jev-latest` model on 2026-09-16. Each point is a confidence bucket plotting mean reported confidence against proxy accuracy, where proxy accuracy is agreement with the modal choice across repeats. Because the proxy scores self-agreement rather than verified correctness, the diagonal indicates consistency between stated confidence and repeat-stable behaviour — not truthfulness.

The expected calibration error of 0.0730 and the Brier score of 0.0252 indicate that reported choice confidence is broadly consistent with the proxy labels, and the mean pairwise noul gap of 0.0050 shows that repeated noul answers on the same state are nearly identical. These claims extend exactly as far as the proxy does. Self-consistency can establish that the model is stable across repeats and that its confidence ordering is internally coherent; it cannot certify that the answers are correct about the world, because the proxy labels are generated by the same model whose calibration is being measured. The benchmark is therefore evidence of calibrated, repeatable decision behaviour under a reproducible proxy — a necessary, not sufficient, condition for deploying these gates on real decisions.

5.4 Interpretation

Four observations tie the measurements back to the design rulings of sec. 4. First, the batching speedups in tbl. 1 confirm that the parallel-sampler semantics translate directly into wall-clock savings: the batched strategy is faster at every configured width, and the gap widens with width exactly as the round-trip accounting predicts. Second, the token-cost ratio below unity at the widest width shows that batching is not merely faster but cheaper, because the state is transmitted once — so the pattern documented in the primary source [TypeSafe AI, 2026c] holds end to end for a third-party client implementation. Third, the pipeline latencies in tbl. 2 sit within the model’s millisecond-scale latency envelope (sec. 3): the pure-logic composition layer adds no measurable latency beyond the network round trip, which is precisely what allows confidence-gated routing to run inline on interactive request paths rather than in a background queue. Fourth, the calibration summary in tbl. 3 supports the calibrated-decision premise of the confidence-gated patterns described in sec. 4 — reported choice confidence aligns with repeat-stable behaviour, and noul answers are stable across repeats — under the self-consistency proxy semantics stated in the Calibration section, and only as far as those semantics extend.

6 Experimental Setup

This section records the environment, configuration, and measurement protocol behind the results in sec. 5, so that the runs can be reproduced bit-for-bit in intent.

6.1 Software environment

All measurements were taken with daf-jev version 0.2.0, running under 3.14.6 on the platform reported as macOS-26.6.2-arm64-arm-64bit-Mach-0. The package targets Python 3.10 or newer and depends, at runtime, only on httpx for transport and pyyaml for configuration parsing; the benchmark scripts additionally use the standard library. The development toolchain is uv-managed, and the unit test suite is executed with pytest under a coverage gate.

The model-level claims in sec. 3 are grounded in a local, hash-manifested snapshot of the TypeSafe documentation (snapshot b79c9cd6008489f1, 108 pages) rather than the live website, so the primary-source basis of this manuscript is itself versioned and verifiable.

6.2 Benchmark configuration

All benchmarks execute against the real jev-latest model with a live API key resolved through the package’s credential precedence chain (injected mapping, then process environment, then the project .env file). Without a key, the scripts print a skip notice and exit successfully — they are benchmarks, not tests, and never run inside the test suite.

- **Batching benchmark** (benchmarks/bench_batching.py) — for each configured batch width (5, 10, 20 questions, as recorded under experiment.batching_n_values in manuscript/config.yaml), it compares one batched call against the same number of sequential single-question calls over a fixed state paragraph and a mixed noul/choice/score question set, repeating each strategy for the configured number of runs and recording wall time and token totals to output/benchmarks/batching_<date>.json.
- **Decision-pattern benchmark** (benchmarks/bench_patterns.py) — runs the composite-score and intent-routing pipelines 6 times each (default recorded under experiment.patterns_runs), reporting mean, median, and tail wall times plus token totals to output/benchmarks/patterns_<date>.json. An asynchronous mode executes the same runs concurrently through AsyncJevClient and compares against the sequential wall time.
- **Calibration benchmark** (benchmarks/bench_calibration.py) — runs 6 short states 5 times each against the jev-latest model, scores choice answers under the self-consistency proxy (agreement with the modal choice across repeats) and noul answers by mean pairwise stability, and writes the expected calibration error, Brier score, per-bucket reliability data, and the mean pairwise noul gap to output/benchmarks/calibration_<date>.json, together with a notes field stating the proxy semantics.

All benchmark scripts take --runs and --model arguments; the defaults are recorded as data under the experiment: block of manuscript/config.yaml, which is the same file the manuscript-variable generator reads — configuration, prose, and figures cannot disagree about the protocol.

6.3 Measurement protocol

Percentiles follow the benchmark helper shared by both scripts: the reported tail statistic is the value at the ceiling-rank position of the ordered sample, computed over the recorded runs rather than a sliding window. Wall time covers the full pipeline — transport, retries (none should occur in a healthy run), parsing, and composition logic — so the numbers in tbl. 2 are conservative upper bounds on what an integrating application would add to its request path. Token totals are read from the response usage records, not estimated.

This manuscript was generated at 2026-09-17T01:36:24Z; the rendered values in sec. 5 correspond to the benchmark JSONs current at that timestamp, and the figure generators resolve “latest” the same way (latest by filename date) so that re-rendering after a new benchmark run updates prose, tables, and figures together.

7 Reproducibility Certification

Every artifact behind this manuscript — figures, tables, token values, and the documentation snapshot the model claims rest on — is regenerable from the repository with the commands below, and every generated value reaches the prose through the token pipeline rather than hand transcription.

7.1 Artifact inventory and hashing

- **Figures.** The six figures of this manuscript are generated into `../figures/` by `src/daf_jev/figures.py` (one `generate_<name>()` function per figure plus `generate_all(out_dir)`), orchestrated by `scripts/generate_figures.py`:

```
uv run python scripts/generate_figures.py          # all figures
uv run python scripts/generate_figures.py --only batching_speedup
```

The data-driven figures read `output/benchmarks/batching_*.json`, `output/benchmarks/patterns_*.json`, and `output/benchmarks/calibration_*.json`, resolving “latest” by filename date; a missing benchmark file is reported as a clear error naming the missing file rather than silently producing an empty chart. The two schematic figures and the parametric confidence-band illustration require no data and no network.

- **Figure registry.** Every figure is recorded in `../figures/figure_registry.json` — six entries, one per figure, each mapping the manuscript’s cross-reference label (`fig:architecture` through `fig:calibration`) to its filename, caption, section, and layout width — so the figure set itself is a versioned data artifact rather than a convention.
- **Calibration benchmark payloads.** The calibration run writes `output/benchmarks/calibration_<date>.json` (per-bucket reliability data, expected calibration error, Brier score, the mean pairwise noul gap, and a `notes` field stating the self-consistency proxy semantics); `benchmarks/bench_calibration.py` regenerates it live, and the reliability figure `../figures/calibration_reliability.png` is rendered from the same payload by `generate_calibration()` in `src/daf_jev/figures.py`.
- **Manuscript variables.** All dynamic values reach the prose as double-brace token placeholders, computed by `src/daf_jev/manuscript_variables.py::generate_variables(project_root)` and written to `output/data/manuscript_variables.json` by the thin orchestrator `scripts/z_generate_manuscript_variables.py`, which then renders substituted copies of every section into `output/manuscript/`. Running in strict mode fails if analysis outputs are missing; the `--allow-draft` flag substitutes draft sentinels instead of failing, for early-stage renders only.
- **Documentation snapshot.** The bundled snapshot of the TypeSafe documentation comprises 108 pages (1014.8 KiB) under `docs/reference/`, scraped on 2026-09-16 and identified as snapshot `b79c9cd6008489f1`. The manifest records a content hash per page plus the snapshot identifier; drift is detectable at any time with:

```
uv run daf-jev docs-verify          # re-hash the tree, exit non-zero on drift
uv run python scripts/scrape_docs.py --check # same check without rewriting
```

7.2 Test suite and coverage

The test suite follows the no-mock convention: unit tests drive the real transport against a real local HTTP server fixture, and live tests hit the real API only when a key is present in the environment (they skip with a notice otherwise). The suite comprises 17 test files — 256 unit tests and 2 live tests — collected with:

```
uv run pytest tests/unit --cov=src          # unit suite under the coverage gate
JEV_API_KEY=... uv run pytest tests/live    # live tests against the real API
```

Measured coverage over the package source stands at 92.16, enforced by the coverage gate configured in `pyproject.toml`. Test and collection counts are computed at variable-generation time by collecting the suite; if collection is unavailable in a given environment, the corresponding values are reported as draft sentinels rather than fabricated.

7.3 Provenance chain

The certification chain is: benchmark scripts write dated JSON payloads → figure generators read those payloads and render `../figures/*.png` → the variable generator reads the same payloads plus `manuscript/config.yaml`, `pyproject.toml`, the test suite, and the docs manifest to compute the token mapping → the injection step substitutes tokens into `output/manuscript/*.md` → the renderer consumes the substituted copies. No numeric fact in this paper has a hand-maintained copy; the environment of record is `macOS-26.6.2-arm64-arm-64bit-Mach-O` under 3.14.6, and the rendered edition is version 0.2.0 of this manuscript, generated at 2026-09-17T01:36:24Z.

8 Scope, Related Work, and Positioning

8.1 Scope and limitations

This manuscript describes an early, private sidecar package, and three caveats bound every claim in it.

Single-model measurements. All benchmarks in sec. 5 were recorded against the `jev-latest` model at the configuration described in sec. 6. The package accepts a per-call model override, but nothing here characterizes how the speedups, token ratios, or latency percentiles transfer across models or across provider-side changes to the same model; re-running the two benchmark scripts against another setting is the intended way to extend the measurements, and the render-time token pipeline picks up the new payloads automatically.

Live-API variance. The measurements are wall-clock observations of a shared remote service. Run-to-run variation from network conditions and provider load is inherent; the percentile protocol in sec. 6 reports medians and tails rather than single samples, but the values should be read as representative of a healthy session, not as service-level guarantees.

Early-access documentation basis. The model-level claims in sec. 3 rest on a hash-manifested snapshot of the vendor’s documentation (snapshot `b79c9cd6008489f1`) rather than on the vendor’s release process. The package documents an early-access surface: wire shapes, model identifiers, and documented patterns can change upstream, and the `docs-verify` command plus a re-scrape (sec. 7) is the mechanism for detecting and absorbing such changes. Behavior not covered by the snapshot is implemented as a stated assumption rather than a verified fact.

Private sidecar status. This project is a private sidecar of the surrounding research infrastructure: it is versioned and licensed (MIT) for its own reproducibility, but it makes no archival deposit claims and carries no DOI.

8.2 Related work

Structured output from generative LLMs. The mainstream approach to software-consumable model answers is constrained decoding: JSON-schema-constrained generation, function calling, and post-hoc validators over generated text. These techniques constrain the *syntax* of free-text generation but inherit its semantics — a well-formed JSON payload can still encode an overconfident or miscalibrated judgment, and nothing in the decoding contract expresses uncertainty as calibrated probability. The decision-model approach of sec. 3 differs at the training level: RLCD optimizes for decisions and calibrated probabilities as the output contract itself [TypeSafe AI, 2026e], rather than for text that a schema hopes to bound.

RLHF versus RLCD. RLHF and its verifiable-reward variants adapt a pretrained model by optimizing text quality against human preferences or checkable rewards. RLCD instead optimizes a different output contract — decisions with calibrated probability distributions — which is what makes the confidence field of a `choice` or `score` answer a trustworthy input to routing policy rather than a stylistic flourish [TypeSafe AI, 2026e,b]. The distinction matters practically: confidence-gated routing (sec. 4) is only sound if the confidence numbers are calibrated, which is a training-time property, not a prompt-time one.

Agent decision layers. Where agent frameworks place the model *in* the control loop — the model chooses its next action — the System One placement is the inverse: code owns control flow and the model supplies atomic judgments at fixed points [TypeSafe AI, 2026d]. Third-party projects in the Jev ecosystem explore this decision-layer space from adjacent angles: an architectural registry for composing Jev-style components [Claburn, 2026], a router that orchestrates decision calls across application workflows [Corvin, 2026], and a standalone evaluation harness for Jev question sets [Morales, 2026]. `daf-jev` is complementary rather than competitive: it contributes a dependency-light, strictly typed *client and composition library* — the tested plumbing layer these efforts can sit on — together with measured evidence for the batching and confidence-routing patterns the ecosystem’s documentation describes qualitatively [TypeSafe AI, 2026c].

9 References

The bibliography lives in `manuscript/references.bib` and is resolved by Pandoc — `--natbib` on the PDF path, `--citeproc` for the other editions. Citation keys used throughout this manuscript are fixed by the project’s research pass: the primary System One source [TypeSafe AI, 2026d], its concept primer [TypeSafe AI, 2026e], the API reference [TypeSafe AI, 2026a], the confidence documentation [TypeSafe AI, 2026b], the decision patterns documentation [TypeSafe AI, 2026c], and the third-party Jev ecosystem works [Claburn, 2026], [Corvin, 2026], and [Morales, 2026]. Every bracketed key in the preceding sections resolves against that file.

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